

Math 208 - Tutorial 2

Mina Moeini
Mina_moeini@sfu.ca

January 14, 2025

1 The Electro- Poly Problem

Electro-Poly is a leading maker of slip rings. They can make some slip rings and buy the remaining units to fulfill the order. Making slip rings requires wiring and harnessing operations. Currently, Electro-Poly has only 10,000 hours of wiring and 5,000 hours of harnessing available. The number of units ordered, resource consumption, and costs are shown in the table below. What should Electro-Poly do?

	Model 1	Model 2	Model 3
Number of slip rings ordered	3,000	2,000	900
Hours of wiring per unit	2	1.5	3
Hours of harnessing per unit	1	2	1
Cost per unit to make	\$50	\$83	\$130
Cost per unit to buy	\$61	\$97	\$145

Resources available:

- Wiring: 10,000 hours
- Harnessing: 5,000 hours

Determine how Electro-Poly should allocate resources and whether to make or buy each model to minimize costs while fulfilling the order.

Solution

Problem Parameters

The problem parameters are the resources required to make or buy a specific slip ring model and the maximum availability of resources.

- Let w_m, h_m be the wiring and harnessing hours required per unit of model m , $m \in M = \{1, 2, 3\}$, where each model corresponds to Model 1, Model 2, and Model 3 respectively.
- Let $W = 10,000$, $H = 5,000$ be the maximum available hours for wiring and harnessing, respectively.
- Let c_m^{make} and c_m^{buy} be the costs to make and buy a unit of model m , respectively.
- Costs for each model:
 - Model 1: $c_1^{\text{make}} = 50$, $c_1^{\text{buy}} = 61$
 - Model 2: $c_2^{\text{make}} = 83$, $c_2^{\text{buy}} = 97$
 - Model 3: $c_3^{\text{make}} = 130$, $c_3^{\text{buy}} = 145$

Problem Decision Variables

The decision variables are:

- x_m : the number of units of model m to make, $m \in M$.
- y_m : the number of units of model m to buy, $m \in M$.

Problem Constraints

The total production and purchase must meet the demand for each model:

$$x_1 + y_1 = 3,000$$

$$x_2 + y_2 = 2,000$$

$$x_3 + y_3 = 900$$

The total resource usage must not exceed the available limits:

$$2x_1 + 1.5x_2 + 3x_3 \leq 10,000 \quad (\text{Wiring constraint})$$

$$1x_1 + 2x_2 + 1x_3 \leq 5,000 \quad (\text{Harnessing constraint})$$

Non-negativity constraints:

$$x_m \geq 0, \quad y_m \geq 0 \quad \forall m \in M$$

Problem Objective

The objective is to minimize the total cost of fulfilling the order. The cost includes both making and buying slip rings:

$$\text{minimize } 50x_1 + 61y_1 + 83x_2 + 97y_2 + 130x_3 + 145y_3$$

The overall model is as follows:

$$\text{Minimize: } 50x_1 + 61y_1 + 83x_2 + 97y_2 + 130x_3 + 145y_3$$

$$\text{Subject to: } x_1 + y_1 = 3,000, \tag{1}$$

$$x_2 + y_2 = 2,000, \tag{2}$$

$$x_3 + y_3 = 900, \tag{3}$$

$$2x_1 + 1.5x_2 + 3x_3 \leq 10,000, \tag{4}$$

$$1x_1 + 2x_2 + 1x_3 \leq 5,000, \tag{5}$$

$$x_m \geq 0, \quad y_m \geq 0 \quad \forall m \in M. \tag{6}$$

2 Make or Buy (From Baker 2nd edition)

A sudden increase in the demand for smoke detectors has left Acme Alarms with insufficient capacity to meet demand. The company has seen monthly demand from its retailers for its electronic and battery-operated detectors rise to 20,000 and 10,000 units, respectively. Acme wishes to continue meeting this demand.

Production Process and Relevant Data

Acme's production process involves three departments: Fabrication, Assembly, and Shipping. The relevant quantitative data on production and prices are summarized below:

Department	Monthly Hours Available	Hours/Unit (Electronic)	Hours/Unit (Battery)
Fabrication	2000	0.15	0.10
Assembly	4200	0.20	0.20
Shipping	2500	0.10	0.15

Cost/Price	Electronic Detector	Battery Detector
Variable Cost/Unit (\$)	18.80	16.00
Retail Price/Unit (\$)	29.50	28.00

The company also has the option to obtain additional units from a subcontractor, who has offered to supply up to 20,000 units per month in any combination of electronic and battery-operated models at a charge of \$21.50 per unit. For this price, the subcontractor will test and ship its models directly to the retailers without using Acme's production process.

Questions

- What is the maximum profit and the corresponding make/buy levels? (This is a planning model, and fractional decisions are acceptable.)
- Trace the effects of increasing Fabrication capacity by 10 percent. How will the optimal make/buy mix change? How will the optimal profit change?

Solution

Problem Parameters

The problem parameters are the monthly available hours for each department, the hours per unit required for each department to create electronics and batteries, the variable cost per unit for both items, the retail price for both items, and the demand:

- $H_d, d \in D = \{f, a, s\}$: The monthly hours available for fabrication, assembly, and shipping.
- $E_d, d \in D$: The hours per unit (electronics) for each department.

- $B_d, d \in D$: The hours per unit (battery) for each department.
- C_e, C_b : The variable cost for electronics and batteries.
- R_e, R_b : The retail price for electronics and batteries.
- S : The price per unit from purchasing electronics or batteries from the subcontractor.
- D_e, D_b : The demand for electronics and batteries.

Problem Decision Variables

The problem decision variables are to select how many electronics and batteries to make or buy:

- Let $x_{e,m}$ and $x_{b,m}$ be the number of electronics and batteries we will make, respectively.
- Let $x_{e,b}$ and $x_{b,b}$ be the number of electronics and batteries we will buy, respectively.

Problem Constraints

The constraints are based on the number of monthly available hours for production, the total demand, and the total number of electronics and battery models we can buy. We denote this as $P_{\max}$. The constraints are as follows:

$$E_d x_{e,m} + B_d x_{b,m} \leq H_d, \quad d \in D, \quad (7)$$

$$x_{e,b} + x_{b,b} \leq P_{\max}, \quad (8)$$

$$x_{e,b} + x_{e,m} = D_e, \quad (9)$$

$$x_{b,b} + x_{b,m} = D_b. \quad (10)$$

Non-Negativity Constraints

$$x_{e,m}, x_{b,m}, x_{e,b}, x_{b,b} \geq 0$$

Problem Objective

The objective is to maximize the overall profit (Revenue - Cost):

$$\text{Maximize: } R_e(x_{e,m} + x_{e,b}) + R_b(x_{b,m} + x_{b,b}) - C_e x_{e,m} - C_b x_{b,m} - S(x_{e,b} + x_{b,b}). \quad (11)$$

The overall model is:

$$\begin{aligned}
\text{Maximize: } & R_e(x_{e,m} + x_{e,b}) + R_b(x_{b,m} + x_{b,b}) - C_e x_{e,m} - C_b x_{b,m} - S(x_{e,b} + x_{b,b}) \\
\text{Subject to: } & E_d x_{e,m} + B_d x_{b,m} \leq H_d, \quad d \in D, \tag{12} \\
& x_{e,b} + x_{b,b} \leq P_{\max}, \tag{13} \\
& x_{e,b} + x_{e,m} = D_e, \tag{14} \\
& x_{b,b} + x_{b,m} = D_b. \tag{15}
\end{aligned}$$

Part a)

The optimal production levels are as follows:

Category	Amount
Electronics (make)	6666.66
Batteries (make)	1000
Electronics (buy)	13 333.33
Batteries (buy)	0

The maximum profit is \$298,000.

By increasing fabrication by 10%, we obtain the following result:

Category	Amount
Electronics (make)	8000
Batteries (make)	10 000
Electronics (buy)	12 000
Batteries (buy)	0

The maximum profit is \$301,600.

More electronics are made.

3 Leasing Warehouse Space (From Baker second edition)

Cox Cable Company needs to lease warehouse storage space for five months at the start of the year. The company knows how much space will be required in each month, and it can purchase a variety of lease contracts to meet these needs. For example, it can purchase one-month leases in each month from January to May. It can also purchase two-month leases starting from January through April, three-month leases from January through March, four-month leases from January and February, or a five-month lease starting in January.

In total, there are 15 possible leases it could use. The company must decide which leases to purchase and how many square feet to purchase on each lease. Since the space requirements differ month-to-month, it may be economical to lease only the amount needed each month on a month-by-month basis. On the other hand, the monthly cost for leasing space for additional months is much less than for the first month, so it may be desirable to lease the maximum amount needed for the entire five months. Another option is an intermediate approach of changing the total amount of space leased (by adding a new lease and/or having an old lease expire) at least once, but not every month. Two or more leases for different terms can begin at the same time.

Space Requirements and Leasing Costs

The space requirements (in square feet) and the leasing costs (in dollars per thousand square feet) are as follows:

Month	Space Requirements (sq ft)	Lease Length (months)	Cost (\$ per 1000 sq ft)
January	15,000	1	280
February	10,000	2	450
March	20,000	3	600
April	5,000	4	730
May	25,000	5	820

Problem Objective

The task is to find a leasing schedule that:

- Provides the necessary amount of space in each month.
- Minimizes the total leasing cost.

Questions

- (a) Determine the optimal leasing schedule and the optimal total cost.

- (b) Trace the effects of increasing the space required for January. How will the leasing schedule change? How will the total cost change?

Solution

This is a covering problem, looking to minimize the total lease cost while having enough storage space.

Problem Parameters

The problem parameter is the lease cost and the monthly requirements. Let c_i be the cost of a 1,000 square foot i -month rental, where $i \in \{1, 2, \dots, 5\}$. Let m_i be the required storage space for month i , where $i \in \{1, 2, \dots, 5\}$.

Problem Decision Variables

The decision variables are the quantity of each rental lease. Let $x_{i,j}$ be the number of rentals from the start of month i to the end of month j , where $i \leq j$ and $i, j \in \{1, 2, \dots, 5\}$.

Problem Constraints

The problem constraint is to ensure that enough space is rented for each month. This can be written as:

$$\sum_{i \leq k \leq j, i, j \in \{1, 2, \dots, 5\}} x_{i,j} \geq m_k, \quad \forall k \in \{1, 2, \dots, 5\}.$$

Non-Negativity Constraints

$$x_{i,j} \geq 0$$

Problem Objective

The objective is to minimize the total cost of the rental. Recall that the cost parameter is per thousand square feet while the required space is in squared feet.

$$\min \frac{1}{1000} \left(\sum_{i=1}^5 c_1 x_{i,i} + \sum_{i=1}^4 c_2 x_{i,i+1} + \sum_{i=1}^3 c_3 x_{i,i+2} + \sum_{i=1}^2 c_4 x_{i,i+3} + \sum_{i=1}^1 c_5 x_{i,i+4} \right)$$

The overall model is:

$$\begin{aligned}
\text{Minimize: } & \frac{1}{1000} \left(\sum_{i=1}^5 c_1 x_{i,i} + \sum_{i=1}^4 c_2 x_{i,i+1} + \sum_{i=1}^3 c_3 x_{i,i+2} + \sum_{i=1}^2 c_4 x_{i,i+3} + \sum_{i=1}^1 c_5 x_{i,i+4} \right) \\
\text{Subject to: } & \sum_{i \leq k \leq j, i, j \in \{1, 2, \dots, 5\}} x_{i,j} \geq m_k, \quad \forall k \in \{1, 2, \dots, 5\}.
\end{aligned} \tag{16}$$

Part a)

Using Excel Solver, the optimal rental schedule is as follows:

- A 15 000 square feet, 5-month rental starting in January.
- A 5000 square feet, 1-month rental in March.
- A 10 000 square feet, 1-month rental in May.

The total optimal cost is \$16, 500.

Part b)

Increasing the January required storage amount to 50 000 square feet resulted in the following rental schedule:

- A 20 000 square feet, 5-month rental starting in January.
- A 30 000 square feet, 1-month rental in January.
- A 5000 square feet, 1-month rental in May.

The total cost changes to \$26, 200.